

Tech Hiring Assignment

Role: Full-Stack Engineer (Fresher) **Time Budget:** 4–6 hours **Submission:** GitHub repository link — mandatory. No GitHub = submission not reviewed.

Before You Begin

- Use any programming language you are comfortable with.
 - Your entire submission — code, writeup, wireframes, use cases/stories, unit tests — must live in a public GitHub repository. Commit as you work.
 - Optimize for efficiency and ease of use when designing user experience.
 - You may use AI tools (ChatGPT, Claude, Copilot, etc.).
 - If you clear this round, your next interview will be based entirely on this submission. We will go through your code, your decisions, and your reasoning line by line.
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The Problem

Hero Cycles is one of the largest bicycle manufacturers in India. You are a fresher engineer joining their small digital team. On Day 1 your manager gives you this brief:

"We sell thousands of cycle configurations — different frames, different gear sets, different tyre types. Our sales team is constantly going back and forth on pricing because part costs change every few months — a tyre priced at ₹200 in January may be ₹230 by December. Right now everything is done on Excel sheets. We need a proper pricing engine — something salesperson(s) can use to manage cycle configurations & parts, and instantly get the total price, broken down by component. Can you build this?"

What You Need to Deliver

Part 1 — Problem Breakdown

Part 2 — Conceptual Solution & Code

Part 3 — Design Sensibility & UI

Submission Checklist

- Please write all the questions you thought of while solving the problem statement
- Please clearly write the assumptions you have made.
- Write Psuedocode for part 2 of the Question.
- Anything else if you think is important for us to know
- Your backend and UI need to be interactive.
- You need to share the Prompts you used for the entire solutioning
 - brainstorming, document writing, coding etc.
- Make sure the readme has clear instructions on how to install and that it works, failing which we may not be able to evaluate your submission.

No GitHub link = submission not reviewed.

For questions, contact [your recruiting contact].